

One Screen, Many Minds: A Smart TV Interface Development for Co-viewing through a Mental Model Framework

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Abstract

Advancements in smart TV technologies have enhanced digital entertainment by providing personalized services to households. However, their role as a group device and the complexities of co-viewing experiences are often overlooked. To better understand user expectations in co-viewing contexts, and create more intuitive and satisfying experiences, we adopt the concept of mental models. By employing a mental model framework, we study co-viewing sessions and conducted semi-structured interviews to capture how groups browse and select content when co-viewing. Our findings reveal five design implications for interface designs including social factors, active user engagement, and adaptive interaction design. Based on these design implications, we propose an interface design prototype informed by four actionable design insights that align with co-viewing usage contexts. By investigating how group dynamics shape content exploration and decision-making, our work delivers practical insights with a visual prototype for designing more satisfying co-viewing experiences on smart TVs.

CCS Concepts

• Human-centered computing;; • Interface design prototyping;; • User studies;;

Keywords

Smart TV, User interface, Prototyping, Mental model, Co-viewing

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1 INTRODUCTION

Smart TV technologies have fundamentally transformed how users interact with digital entertainment devices, situating personalized

services at the forefront of modern digital experiences. In fact, smart TVs now appear in nearly 79% of U.S. homes [30], delivering richer content and advanced features right into living rooms [9, 24]. However, the fundamentals of smart TV usage have not dramatically changed. Like their predecessors, smart TVs still anchor our living rooms to serve as an entertainment hub where families and friends gather. While watching TV can be viewed as a solitary activity, it is often enjoyed in groups, providing topics for conversation and fostering a sense of togetherness [23, 28]. Whether at home with loved ones or in public venues like sports bars, it remains integral to TV consumption [16]. Recognizing the social nature of smart TVs calls for design that accounts for the complexities of group interactions. After all, smart TVs are considered group devices [5, 19, 20, 36], entailing interface designs that acknowledge co-viewing dynamics for a satisfying and intuitive user experience for co-viewing.

Focusing on a personalized smart TV experience by leveraging individual data can effectively cater to solitary preferences, yet it often struggles to support the browsing process when multiple viewers are involved [4]. Though some studies have aimed to identify individuals for the group recommendation interface [3, 12, 22, 36], or integrate secondary devices for multi-screen usage [13, 25], they often solely focus on the algorithmic implementation into the interface, leaving the nuanced group dynamics and decision-making processes to capture the complexity of co-viewing.

Moreover, beyond having a recommender interface, engaging users in exploration can be essential to a satisfying co-viewing experience. While it is true that smart TVs are lean-back devices with passive viewers when consuming the content [5], browsing for content in a co-viewing context may require active involvement for a better outcome. Research on consensus models shows that active discussion and adaptable opinions among group members lead to more satisfying decisions [31]. Therefore, designing interfaces that encourage curiosity and collaboration can help make the progress more engaging and enjoyable for everyone.

To better capture these dynamics, we turn to the concept of mental models formed in people's minds of reality which can be used to predict events in the real world [10]. They are considered cognitive representations that users form to understand the physical world and constructs that can explain human behavior [18, 27]. Prior work demonstrated that mental models influence users' comprehension, problem-solving, and consensus-building [32]. In the realm of interface designs, examining mental models as conceptual frameworks for interpreting and predicting user interactions can help explain how the users form and deploy varied knowledge representations so that designers can accommodate the differences

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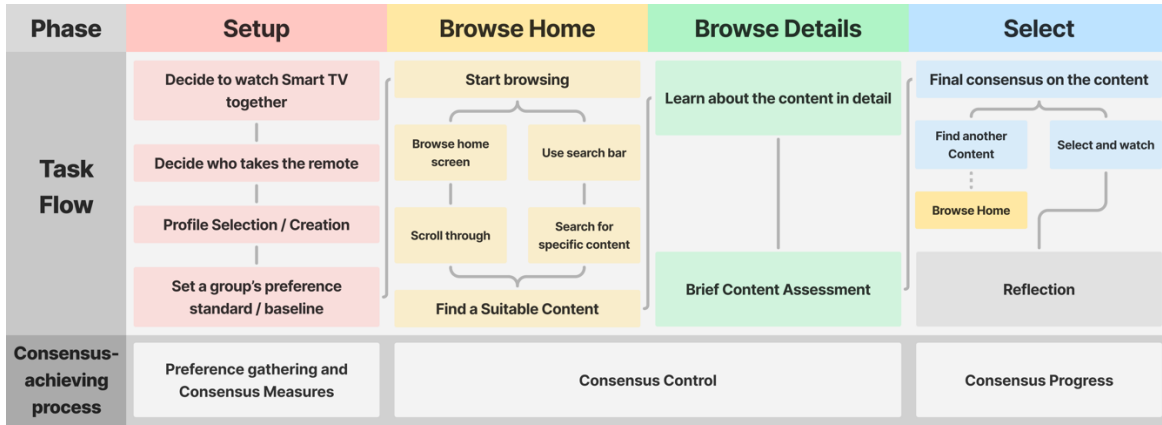


Figure 1: Task Structure of Browsing a Smart TV in a Co-viewing Context

[6]. Thus, this approach can not only help designers understand the frameworks that guide group content browsing and decision-making but also assist in managing users' expectations and levels of satisfaction with the product which enables the interfaces that better align with end-users' needs [1].

This study aims to suggest an interface for smart TVs in a co-viewing context by employing a mental model approach to explore the intricacies of co-viewing scenarios. From these findings, we derive design implications reflecting common co-viewing challenges, which then shape a prototype interface intended to enhance the user experience in co-viewing contexts. We pose the following research questions to guide our work:

(RQ1) How does the mental model of content browsing for co-viewing differ from individual content browsing?

(RQ2) How could users be more engaged in finding new content?

(RQ3) What are the challenges users face when reaching a consensus in a co-viewing context?

2 QUALITATIVE DATA COLLECTION FOR THE MENTAL MODEL

This section outlines the qualitative data collection process prior to developing the mental model, including participant observation of users' real-world browsing behaviors and semi-structured interviews exploring participants' rationales. Observations and interviews were transcribed and coded to uncover patterns in reasoning, negotiation, and decision-making.

2.1 Co-viewing Task Structure and Semi-structured Interview Protocol

We incorporated a consensus-building process into the task structure to reflect the co-viewing context which includes a mechanism to measure, maintain, and enhance agreement as group members interact [7]. Figure 1 shows four phases in the smart TV co-viewing task: *Setup*, *Browse Home*, *Browse Details*, and *Select*. First, a group agrees on the baseline of shared preferences during *Setup*, then moves on to explore available content through browsing, searching, and discussing options during the *Browse Home* phase. As they

progress to *Browse Details*, they dive deeper into potential selections, weighing how well each title meets the group's standards. Finally, during the *Select* phase, the group decides whether to finalize a choice or loop back for a higher consensus. After the task structure was established, the semi-structured interview protocol was built around it. The questions were aligned with the established task phases to ensure contextual relevance, and only the relevant questions were selected from the pre-organized list (see Table 2 in Appendix A.1) to accommodate variability in participants' usage patterns.

2.2 Participants

We recruited 11 groups of participants in pairs with the inclusion criteria of having a close relationship to ensure authentic real-world behavior. Specifically, the groups comprised seven friend groups, three family groups, and one romantic couple. The groups had an average age of 28.2 years ($SD=7.6$), with an equal split of males ($N=11$) and females ($N=11$). All participants were regular users of smart TVs in their homes. The task was video recorded to enable subsequent behavioral analysis, and the entire procedure, including the semi-structured interview, took 60 minutes to complete per group.

2.3 Environment Setup

The study was designed to resemble a typical living room setting to ensure realistic observation and data collection, facilitating the natural behaviors of participants. The device utilized was the Samsung 2023 QLED 4K QCE1 75-inch (189 cm) TV, paired with Google Chromecast TV as the smart TV platform, chosen for its Android-based interface familiar to most users. The setup included comfortable furniture to create a relaxing atmosphere, encouraging participants to express their thoughts and feelings freely.

2.4 User Observation and Semi-structured Interview Procedure

A brief introduction was followed by a pre-survey to collect information on participants' general habits regarding smart TV platform usage. This preliminary step also served to establish rapport with

participants, ensuring they felt comfortable and understood that there were no correct answers to the questions. Then, they were assigned two separate profiles to simulate personalized content preferences. These profiles utilized the “Content Preference” feature of Google Chromecast TV to tailor their preferences, which allowed the pseudo-user accounts to align with their personal preferences, enhancing the immersive experience and authenticity of interactions within the simulated environment.

The participants were then provided with a common scenario (e.g., a casual movie night for friend pairs or a typical home viewing for families) to help with immersion. Each group was instructed to explore the smart TV interface and ultimately selecting the content they wished to watch together. During this process, participants were free to navigate and interact with the recommended content on the smart TV home interface. After the selection, they were instructed to go back to home screen. Their task performance was video-recorded to capture decision-making behavior processes and group dynamics. Researchers refrained from providing specific instructions during the task to preserve natural behavior.

Upon completion of the task, participants took part in a semi-structured group interview designed to contextualize and elaborate on their observed behaviors. The interviews, which were recorded and later transcribed, aimed to collect qualitative data on participants’ decision-making processes and their reflections during content selection. Open-ended questions were employed to elicit a broad range of feedback, allowing participants to provide detailed insights into their experiences and behaviors during the task.

3 MENTAL MODEL DEVELOPMENT

We developed a mental model for co-viewing experiences on a smart TV using Indi Young’s framework, focusing on capturing the underlying behaviors, intentions, and thought processes of users [35]. It involves a structured, three-step method: first, we combed through transcripts to identify discrete tasks; second, we clustered these tasks to reveal underlying patterns; and finally, we arranged these clusters into a coherent visual model. By exploring not only what users do but also why they do it, this approach uncovers design strategies toward more intuitive and engaging co-viewing interfaces.

3.1 Comb for Tasks: Distilling into Action-oriented Tasks

The first step, “combing” through transcripts, entails a line-by-line review of the raw interview data and observation videos. Each group’s narrative and behaviors are dissected to isolate the actions, decisions, and emotional responses they exhibited while browsing content together on a smart TV to form a total of 197 supporting quotes. Then, similar supporting quotes were grouped together and distilled into concise, action-oriented tasks, each described in a verb+noun phrase without gerunds. For example, one of the supporting quotes is, “Hmm, a philosophical move, and it’s getting really good reviews,” in response to a partner’s novel suggestion, which we coded as “Consider partner’s recommendation for a new experience.” The results of the combed tasks and their supporting quotes are listed in Appendix A.2 (Table 3).

3.2 Find Patterns: Grouping Tasks into Task Towers

After tasks were extracted, the next step involves identifying patterns by grouping related tasks together. These tasks were organized into clusters known as task towers. These towers share a common purpose, respond to a similar trigger, or reflect a particular emotional state. This allowed us to uncover meaningful relationships and themes that might otherwise remain hidden within raw data. This pattern-recognition stage helps clarify the cognitive frameworks that guide user decision-making, illuminating how various behaviors fit together in the broader co-viewing context.

3.3 Finalize the Mental Model: Defining High-level Themes in Mental Spaces

Mental spaces serve as abstract conceptual domains that capture high-level patterns of user activities, transcending individual tasks to focus on underlying motivations, intentions, and behaviors. In developing these mental spaces, we began with a bottom-up analysis: we first identified discrete user actions from transcripts, grouped them into task towers to highlight related behaviors, and then examined these towers to uncover overarching thematic structures. During this process, it became clear that participants’ actions did not strictly follow the chronological sequence suggested by the task structure in Figure 1. Instead, their behaviors naturally aligned with three conceptual areas of *Goal Setting*, *Item Identification*, and *Item Evaluation & Selection*, which we derived from the group decision-making outlined in [15].

As shown in Figure 2, each mental space is represented as a series of task towers. Within each task tower, tasks are stacked vertically. Tasks positioned higher in a task tower have appeared more frequently in the collected data, indicating their relative importance or commonality among participants. Additionally, certain tasks are bordered and numbered with saturated colors to map directly to the findings below. Together, these mental spaces, task towers, and tasks provide a structured visual representation of the co-viewing experience. The mental model is textualized in Appendix A.3 (Table 4).

4 DESIGN IMPLICATIONS

4.1 Social and Contextual Influences in Co-viewing

RQ1: How does the mental model of content browsing for co-viewing differ from individual content browsing? Findings tied to this research question are Finding #1, #2, #3, and #4 in Table 1. We observed that most participant groups (6 out of 11) gravitate toward profiles with narrower variance to ensure minimal friction when deciding what to watch. As participant groups stated, “It will be easier to yield to find content on this profile since I have a comprehensive taste in movies” (P7). Though it may appear that users are simply picking the most compatible profile, the underlying motive is inherently social: different relationships and situations yield different content needs, motivating participants to align the profile with the context at hand. Once the browsing task started, most participants settled on popular content that is easily recognizable,

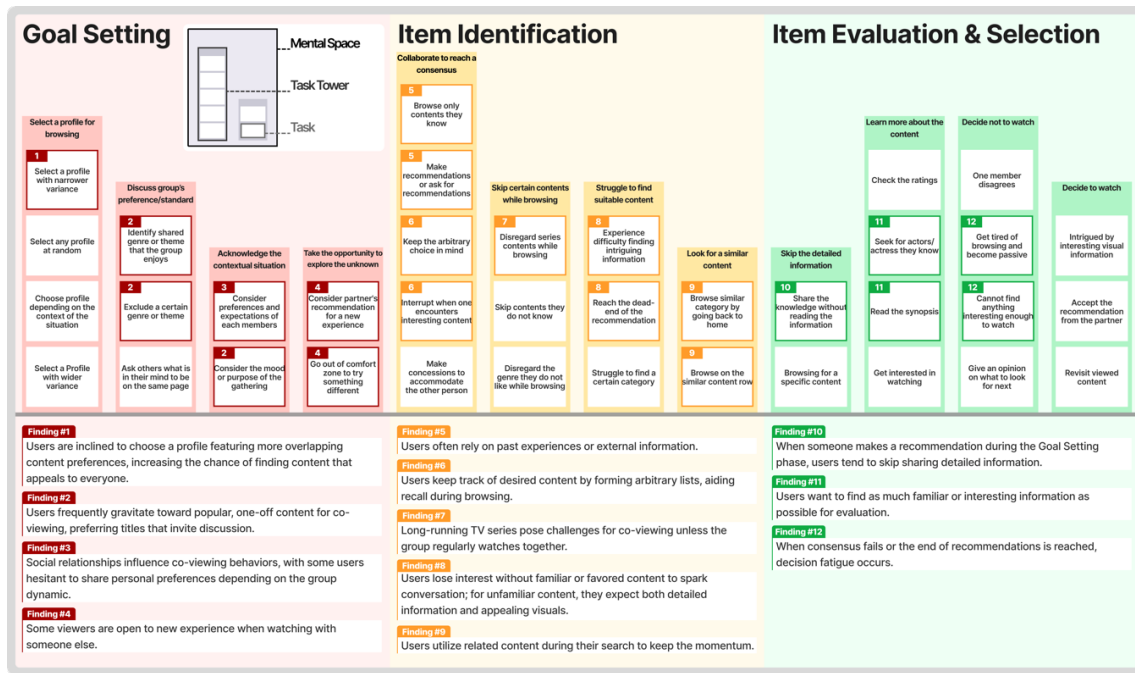


Figure 2: Visualization of User Mental Model of Content Browsing in Smart TV Co-viewing and Corresponding Findings

Table 1: Findings from User Mental Model Categorized by Research Questions

Research Questions	Findings
RQ1	#1 Users are inclined to choose a profile featuring more overlapping content preferences, increasing the chance of finding content that appeals to everyone. #2 Users gravitate toward popular, one-off content for co-viewing, preferring titles that invite discussion. #3 Consider the preferences and expectations of the group members #4 Go out of comfort zone to try something different
RQ2	#5 Users often rely on past experiences or external information to guide their browsing. #6 Users keep track of desired content by forming an arbitrary list, aiding recall during browsing. #8 Users lose interest without familiar or favored content to spark conversation; for unfamiliar content, they expect both detailed information and appealing visuals. #10 When one makes a recommendation during the Goal Setting, users tend to overlook detailed information.
RQ3	#11 Users want to find as much familiar or interesting information as possible for evaluation. #7 Long-running TV series pose challenges for co-viewing unless the group regularly watches together. #9 Browse similar category by going back to home #12 When consensus fails or the end of recommendations is reached, decision fatigue occurs.

or one has already watched to minimize the failure. Also, participants often hesitated to voice personal favorites if they believed those preferences conflicted with group expectations. For instance, many groups (7 out of 11) reported, “TV showed some content I don’t want my family to see” (P2). This self-censorship stems from perceived social pressure in co-viewing contexts. Another participant described balancing safety and novelty in their choices by stating, “I want to dare myself to watch something scarier since I am with someone else right now” (P10). This inclination to step out of one’s comfort zone appears more readily when sharing the experience with someone of different tastes. Collectively, these

behaviors underscore the importance of “safe bets” (i.e., popular or trending titles) and more serendipitous curation that caters to at least one group member’s interests. Thus, the following design implication emerges: (D1) *The design should consider social factors to help with the Goal Setting phase in the co-viewing context.*

4.2 User Engagement Through User Behavior Alignment and Interaction

RQ2: How could users be more engaged in finding new content? Findings tied to this research question are Finding #5, #6, #8, #10, and #11 in Table 1. The interview found that many participants

relied on external cues, such as reviews, influencer recommendations, or previous viewing experiences, to guide initial discovery. Many participants (8 out of 11) stated, “I heard this movie was good from an outside source,” (P22) reflecting how external endorsements can bolster trust and reduce uncertainty about unfamiliar titles. Without such cues, new or lesser-known options risked being dismissed prematurely, suggesting that interfaces should offer contextual clues (e.g., brief synopses or visuals) to ensure novel content does not appear too alien. Moreover, social dynamics played a significant role in amplifying interest, which sparked immediate group discussion and collective excitement. Another way participants stayed engaged was by maintaining a mental or written list of potential titles, as participants were setting “a mental priority, browse further, and then go back,” (P10) allowing them to revisit interesting finds later without losing track during the browsing process. Overall, user engagement hinged on a balance of comfort and novelty, supplemented by memory aids and social cues that mitigate unfamiliarity. These insights translate into the following design implications: (D2) *The design should reflect users’ real-world behaviors to maintain engagement while minimizing cognitive overload;* (D4) *The design should help lower the psychological barriers by facilitating social involvement.*

4.3 Management of Decision-Making Process via Refinement to Enhance Engagement

RQ3: What are the challenges users face when reaching a consensus in a co-viewing context? Findings tied to this research question are Finding #7, #9, and #12 in Table 1. Participants struggled to maintain agreement when the process became prolonged or when content options grew unwieldy. A couple of groups encapsulated this challenge by saying, “I think we should look for a completely different type of content” (P13, P14). Such statements and behavior underscored how readily users reverted to the initial *Goal Setting* phase whenever dissatisfaction arose. Despite some platforms offering features for related content, only two participant groups consistently used these features to keep exploration momentum. Furthermore, some groups (3 out of 11) showed frustration, and “decision fatigue” frequently set in when the group cycled through a range of recommendations without finding common ground. Once fatigue took hold, enthusiasm waned, and the process stalled. Taken together, these findings emphasize that co-viewing interfaces must both streamline the realignment of user preferences and alleviate the mental strain that emerges from stalled negotiations. Thus, the design implications are: (D3) *The design should guide users to refine their goal during the item identification phase;* (D5) *The design should implement solutions to reduce fatigue and sustain engagement throughout the consensus-building process.*

5 SMART TV INTERFACE PROPOSAL FOR CO-VIEWING

5.1 Overview

Based on the design implications from our research, we present a smart TV interface prototype (see Figure 3) for co-viewing. Unlike conventional smart TV interfaces, this prototype is designed to

actively address the distinct needs and behaviors of co-viewing users. Specifically, we seek to satisfy the following design insights:

(Insight #1) Enable dedicated features for co-viewing experience. (Derived from D1)

(Insight #2) Support user engagement by reflecting user behaviors. (Derived From D2)

(Insight #3) Provide adaptive prompts to support the refinement of goals. (Derived From D3)

(Insight #4) Set interactive design strategies for user engagement and help overcome psychological barriers. (Derived from D4 and D5)

5.2 Interface Design Development

5.2.1 (Insight #1) Dedicate Groups into a Specific Mode for Co-viewing Context. Current smart TV systems adopt a “one-size-fits-all” approach, offering limited differentiation beyond tailored recommendations for individual users. The prototype introduces a “Group Mode” where the system merges each user’s data into a single group profile. In the profile merging approach, each individual’s preferences are first transformed into a universal feature vector, then combined via a total distance minimization to aggregate the selected individual profiles [36]. Users can choose which profiles to merge (Figure 3a), and an optional setting is available to filter out explicit content to avoid uncomfortable situations (Figure 3b). The prototype employs labeled rows that communicate why certain content to enhance user clarity and trust [14]. The rows are curated to mirror what group users look for during a co-viewing session which includes rows like “Top Picks for Everyone” to ensure the group’s aggregated preference (Figure 3c), “Short & Sweet” for one-off content, and “Conversation-Starters” to curate content likely to spark a group discussion. The prototype also puts a mutual awareness in social software where users see an aggregated view of the group’s collective tastes rather than overly specific details, thus balancing privacy and transparency [8]. The “One of us enjoy” row is an example of mutual awareness where individual user preferences are acknowledged to increase perceived personalization.

5.2.2 (Insight #2) Employ Design Strategies that Align with Real-World Behaviors. Unlike existing interfaces, the prototype provides a “shopping cart” feature to enhance user engagement and reflect how co-viewers select content (Figure 3d). As seen during user study, this feature allows the group to add interesting titles easily, instead of always relying on collective memory or bookmarks. It gives them easy access to a visual repository of potential choices, helping users avoid losing track of potentially appealing content and keeping the momentum alive [29]. Moreover, building on the notion that visibility and clarity of choices enhance decision confidence [17], each item placed in the cart informs the interface about emergent group preferences. Two additional content rows are introduced to the interface on these user-driven signals. First, the “Surprises for Everyone” row provides a layer of unexpected yet relevant suggestions (Figure 3e), aligning with the findings from the interview where users are keen on new experiences. The concept of serendipitous discovery in recommendation systems has been shown to boost exploration, enjoyment, and overall satisfaction by exposing users to items they might not have otherwise encountered [2, 26]. Second, the “Because It’s in Our Cart” row leverages

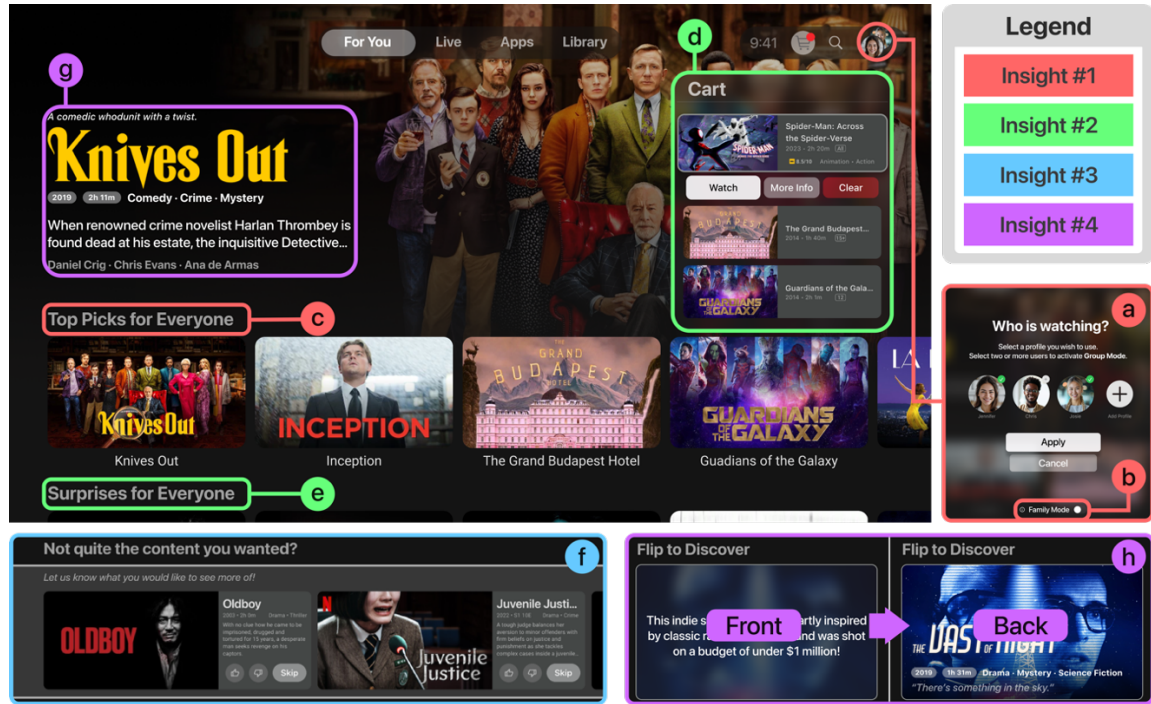


Figure 3: Overview of the Group Mode interface design. Features for Insight #1 include the “Group Mode” (a) with an explicit content filter (b) and content rows for group contexts (c). Features for Insight #2 include a shopping cart (d) and a serendipitous content row (e). Features for Insight #3 include content curation refinement (f). Features for Insight #4 include same-depth content information (g) and trivia cards (h).

contextual similarity, thereby reducing friction in finding adjacent genres or themes.

5.2.3 (Insight #3) Refine Group’s Browsing Goal with an Adaptive Prompt. Current interfaces lack mechanisms to identify when users stall during decision-making. The prototype addresses this by introducing a gentle adaptive prompt whenever participants frequently back out of detail pages in quick succession (Figure 3f). Specifically, after noticing three consecutive “backs” from different show pages, the interface displays a row on the interface asking if users wish to refine or clarify their current objective. This design choice resonates with the principle of timely feedback loops in collaborative tasks, as demonstrated by Kim et al. [21] in their Meeting Mediator system, which demonstrated how real-time sociometric feedback can enhance group performance and dynamics. By analogy, our approach offers immediate contextual cues with refinement options that encourage viewers to reflect on or refine their objectives during browsing, rather than drifting aimlessly. Moreover, by providing clear and timely prompts, the interface can foster users’ acceptance and overall trust in the system [11].

5.2.4 (Insight #4) Reduce Novelty Barriers with Interactive Explorations and Visual Previews. The social context in co-viewing situations creates opportunities for users to explore the unknown, driven by the influences and preferences of others. To support their exploration, the prototype displays detailed visuals and supplementary information at the same “browsing depth” (Figure 3g) since

flattening the interface hierarchy can reduce user effort and encourage broader exploration [34]. Relevant metadata and imagery are revealed rather than forcing users into a deeper page, reducing navigational overhead and offering content previews in place can boost user curiosity and exploratory behavior. Consequently, users are more inclined to take a closer look at recommendations they might otherwise skip, which supports broader engagement with novel or lesser-known options. To further lower the barriers to exploring new content, it includes a “Flip to Discover” trivia card row (Figure 3h) that harnesses social gamification to foster group engagement [8]. The cards evoke a sense of playfulness by layering behind-the-scenes details and fun facts [33]. It places playful treats directly in the user’s browsing interface, encouraging co-viewers to share spontaneous reactions and curious commentary.

6 CONCLUSION

This study investigated the complexities of co-viewing on smart TVs by highlighting how group interaction, consensus-building, and shared preferences differ markedly from solitary viewing. We detailed user behaviors and identified design implications that informed our prototype interface through a user-centered mental model approach, which prioritizes social context, adaptive exploration, and consensus-focused features. Although our findings demonstrate promising avenues for accommodating multiple users, the prototype is still under development. In essence, the interface aims to seamlessly merge individual tastes and social cues into a

shared browsing experience. In future work, we plan to broaden our participant pool to include more diverse user groups and conduct deeper empirical validation of usability and satisfaction, thereby refining these co-viewing design strategies.

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A APPENDICES

A.1 Interview Questionnaires of Semi-structured Interview

Table 2: Interview questions aligned to task flow of users

Phase	Task Flow	Questions
Setup	Decide to watch Smart TV together Decide who takes the remote	
	Profile selection / Creation	- How would you describe your personal content preferences? In what ways do you think they are similar or different from the other person's preferences? - What made you decide to start searching for content this particular way (for example, using a specific account, profile, or method)? - If this method/account isn't perfectly aligned with your preferences, what led you to use it anyway?
	Set a group's preference standard/baseline Start browsing	- When you pick something to watch, what factors do you usually consider? Do you and the other person share your preferences out loud, or do you decide in another way?
Browse Home	Use search bar / Search for specific content Browse home screen / Scroll through Find a suitable content	- What made you want to look elsewhere for content or switch apps? Were you not finding what you wanted in your initial search, or did you have something specific in mind? - I notice you scroll up and down (or browse) quite a bit. Are you searching for a particular type of content or category? What makes you keep looking? - What kinds of things did you say to each other while deciding what to watch? Did you make suggestions, comment on things you saw, or consider each other's typical favorites? - How did you both participate in finding content to watch? Did one person mostly handle the controls while the other offered ideas, or did you take turns? - (If relevant) Is there someone who usually makes the final call, and why? - Why did you choose to split the roles this way? Did it feel like the choice of content was influenced by whoever was leading?
	Learn about the content in detail	- If it didn't feel biased, how did you make sure both of you were satisfied with the choice? - What details about the show or movie made you get interested? - Is there any information you wish you had, or something you saw that didn't really matter? - You picked this without looking at many details. What made you comfortable doing that? Were you already familiar with it, or did you have enough information from elsewhere?
Select	Brief content assessment	- How did you land on this choice to watch together? Did popularity, specific preferences, or someone's suggestion play a big role?
	Final consensus on the content	- When you don't immediately agree on what to watch, how do you usually handle that? For example, do you avoid certain genres, focus on a favorite actor, or pick something you both recognize? - Which piece of information tipped the scales in making this final choice? (For instance, was it the popularity, ratings, cast, plot, or a friend's recommendation?)
	Find another content Select and watch Reflection	- While exploring, you did [X action]. Can you walk me through what prompted you to do that? - How do you feel about having your past viewing habits or preferences visible to others you watch with? Does it change depending on who that person is? - Do you notice any differences in the kinds of shows or movies you pick when you're by yourself compared to when you're with someone else? - Do you feel your content preferences shift based on who you're watching with (e.g., a partner, friend, family member)? Can you share an example? - Did you end up watching something because of certain explanations, details, or recommendations you came across? - When deciding what to watch, what information do you wish you had that isn't available? - What do you find most helpful to know in advance? - Do you have any other thoughts about how today's experience of finding content could be improved or made more enjoyable?

A.2 Supporting Quotes for Tasks

Table 3: Supporting quotes for tasks

Mental Space Task	Supporting Quotes	Count
Mental Space: Item Identification		
1.1 Select a profile with narrower variance	It will be easier to yield to find content on this profile since I have a comprehensive taste in movies	6
1.2 Select any profile at random	It didn't matter which profile we used since we shared a similar preference.	3
1.3 Choose profile depending on the context of the situation	If we are watching at home, the profile with more movies suits the situation better.	1
1.4 Select a Profile with wider variance	I am willing to leave my comfort zone a little when watching with someone else.	1
2.1 Identify shared genres or themes that the group enjoys	I had a handful of contents I wanted to watch in my head; I prefer content that has a lot to talk about; Because she hates scary movies, we tried to avoid them.	6
2.2 Exclude a certain genre or theme	Because she hates scary movies, we tried to avoid them; We should avoid series contents because it's too long; Sequels might have a continuous storyline, and I don't want to have to go back and watch them all.	5
2.3 Ask others what is in their mind to be on the same page	What do you want to watch right now?	4
3.1 Consider the preferences and expectations of the group members	I prefer watching something that can start a discussion; There are some things I don't want my family to see, and I'm not sure I'd be comfortable with R-rated content in that situation.	12
3.2 Consider the mood or purpose of the gathering	I prefer watching jolly and lighthearted contents when watching with a friend; We assumed we had limited time because we were on a vacation trip scenario; We usually binge watch drama series at home, but given the situation, one-off content seemed appropriate	9
4.1 Consider partner's recommendation for a new experience	Hmm, a philosophical/artistic movie, and it's getting really good reviews; If this is not as scary as the other one, then I think I can handle it.	7
4.2 Willing to go out of comfort zone	I want to dare myself to watch something scarier since I am with someone else right now.	2
Mental Space: Item Identification		
1.1 Browse only contents they know	The only content that catches my eyes are those that I know of; I heard this movie was good from an outside source; My friends said this movie is incredible, so I was just kind of went on board with it	18
1.2 Make recommendations or ask for recommendations	I came across this content online, and many people recommend it; I'm okay with watching this again. I think you will like this movie as well; I feel like watching this movie today. How about you?	12
1.3 Keep the arbitrary choice in mind	I set a mental priority, browse further, and then go back if nothing is higher than your priority.	11
1.4 Interrupt when one encounters an interesting content	Have you seen this movie/show?	6
1.5 Make concessions to accommodate the other person	I listen to what others want to watch first. It makes it easier to make decisions; I usually go with what the other person says they want to see minus what I don't like.	6
2.1 Disregard series contents while browsing	We should just watch a one-off content. I don't want the flow to break while watching a long series.	11
2.2 Skip contents they do not know	I tend to ignore things I've never heard of; Contents are presented in an uninteresting way, so I've lost interest in exploring new things.	7
2.3 Disregard the genre they do not like while browsing	Because we both hate horror genre, we are disregarding all horror movies; I hate scary stuff. So, I asked my partner to leave them out of our choice.	3
3.1 Experience difficulty finding intriguing information	The information shown are not helpful. The only information that I can go off is the thumbnail; Once we browse through everything on the home screen, there is no way to refresh the contents.	4
3.2 Reach the dead-end of the recommendation	It's the end of the screen already; Let's go back to the top.	3
3.3 Struggle to find a certain category	It's hard to find the genre I want because I have to go down and look for it because there's no category function.	2
4.1 Browse on the similar contents row	Because we set a baseline of what to look for, it felt natural to browse for a similar content	3
4.2 Go back and browse on the home screen	I browsed through the 'similar to this' row to see if I could find something better than what I have selected	2
Mental Space: Item Evaluation & Selection		
1.1 Share the knowledge without reading the information	This is the guy on the thumbnail is from the other movie we watched; It's based on a famous comic.	6
1.2 Browsing for a specific content	Because we've decided to watch this content beforehand, we didn't read any information shown on screen; What I know about the content was more important than what I saw on the screen.	2
2.1 Check the ratings	Ratings don't tell me if the content is right for us; I liked this movie. Why is the rating so low; The ratings were the only information I can make a reference on how good the movie is.	8
2.2 Seek for actors/actress they know	I didn't know this actor was in this movie. We might watch this.	7
2.3 Gain interest for the content	Visual cues are very important in grabbing our attention; The trailer playing in the background intrigued me to check the content.	7
2.4 Check the synopsis	I would like to know the general storyline of the content.	3
3.1 One member disagrees	It is fair to embrace the disagreements to ultimately reach a consensus; If we disagree on this, I think we should look for a completely different type of content.; We watched it too many times.	5
3.2 Get tired of browsing and become passive	I don't know what to pick, you choose.	4
3.3 Cannot find anything interesting enough to watch	Is this the end of the given contents? There isn't much to go off from.	3
3.4 Give an opinion on what to look for next	Let's try the [type of genre] next.	3
4.1 Intrigued by interesting visual information	Trailers and thumbnails played a big part in deciding what to watch.	9
4.2 Accept the recommendation from the partner	If he/she recommends, I want to give it a shot.	6
4.3 Revisit viewed content	We forgot most of the storyline but we both enjoyed this movie, so it is worth a revisit.	6

A.3 Mental Model of Content Browsing in Smart TV for Co-viewing

Table 4: Mental Model of Content Browsing in Smart TV for Co-viewing

Mental Space	Task Tower	Task	Findings
Goal Setting	1. Select a profile for browsing	1.1 Select a profile with narrower variance	#1 Users are inclined to choose a profile featuring more overlapping content preferences, increasing the chance of finding content that appeals to everyone.
		1.2 Select any profile at random	
		1.3 Choose profile depending on the context of the situation	
		1.4 Select a Profile with wider variance	
	2. Discuss group's preference or standard	2.1 Identify shared genres or themes that the group enjoys	#2 Users gravitate toward popular, one-off content for co-viewing, preferring titles that invite discussion. #3 Consider the preferences and expectations of the group members
		2.2 Exclude a certain genre or themes	
		3.1 Consider the mood or purpose of the gathering	
		3.2 Consider the preferences and expectations of the group members	
	3. Acknowledge the contextual situation	4.1 Go out of comfort zone to try something different	#4 Go out of comfort zone to try something different #4 Go out of comfort zone to try something different
		4.2 Consider partner's recommendation to widen the selection	
		1.1 Make concessions to accommodate the other person	
		1.2 Interrupt when one encounters an interesting content	
Item Identification	1. Collaborate to reach consensus	1.3 Browse only contents they know	#6 Users keep track of desired content by forming mental lists, aiding recall during browsing. #5 Users often rely on past experiences or external information to guide their browsing. #5 Users often rely on past experiences or external information to guide their browsing. #6 Users keep track of desired content by forming arbitrary lists, aiding recall during browsing. #6 Users keep track of desired content by forming arbitrary lists, aiding recall during browsing.
		1.4 Make recommendations or ask for recommendations	
		1.5 Keep the arbitrary choice in mind	
		1.6 Figure out what partner has in mind to be on the same page	
	2. Skip certain contents while browsing	2.1 Disregard the genre they do not like while browsing	#6 Users keep track of desired content by forming arbitrary lists, aiding recall during browsing. #7 Long-running TV series pose challenges for co-viewing unless the group regularly watches together.
		2.2 Disregard series contents while browsing	
		2.3 Skip contents they do not know	
		3.1 Reach the dead-end of the recommendation	
	3. Struggle to spot a suitable content	3.2 Experience difficulty finding intriguing information	#8 Users lose interest without familiar or favored content to spark conversation; for unfamiliar content, they expect both detailed information and appealing visuals. #8 Users lose interest without familiar or favored content to spark conversation; for unfamiliar content, they expect both detailed information and appealing visuals.
		3.3 Struggle to find a certain category	
		4.1 Browse similar category by going back to home	
		4.2 Browse on the similar content row	
Item Evaluation & Selection	1. Skip the detail information	1.1 Share the knowledge without reading the information	#9 Browse similar category by going back to home #9 Browse similar category by going back to home #10 When one makes a recommendation during the Goal Setting, users tend to overlook detailed information.
		1.2 Browsing for a specific content	
		2.1 Seek for well-known actor/actress	
		2.2 Read the synopsis	
	2. Learn more about the content	2.3 Check the ratings	#11 Users want to find as much familiar or interesting information as possible for evaluation. #11 Users want to find as much familiar or interesting information as possible for evaluation.
		2.4 Get interested in watching	
		3.1 Give an opinion on what to look for next	
		3.2 One member disagrees	
	3. Decide not to watch	3.3 Get tired of browsing and become passive	#12 When consensus fails or the end of recommendations is reached, decision fatigue occurs. #12 When consensus fails or the end of recommendations is reached, decision fatigue occurs.
		3.4 Cannot find anything interesting enough to watch	
		4.1 Intrigued by interesting visual information	
		4.2 Revisit viewed content	
	4. Decide to watch	4.3 Accept the recommendation from the partner	